

POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year	: 2025
Programme: BE		Full Marks	: 100
Course: Algebra and Geometry		Pass Marks	: 45
		Time	: 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

- Define consistence and inconsistency of a system of linear equations. Check consistence and inconsistency of given system of linear equations: $5x + 3y + 7z = 4$, $3x + 26y + 2z = 9$, $7x + 2y + 10z = 5$ and solve. [7]
 - Find the eigenvalues and corresponding eigenvectors of the matrix. [8]

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

- Define linear transformation from one space to other. A transformation $T: R^2 \rightarrow R^2$ defined by $T(x, y) = (x + y, xy)$. Test the linearity of T. Does the vectors $(1, 2, 1)$, $(2, 1, 0)$, $(1, -1, 2)$ linearly independent? [7]
 - Find the dual of the primal and solve by using simplex method. Minimize $Z = 8x_1 + 9x_2$ subject to $x_1 + 3x_2 \geq 4$, $2x_1 + x_2 \geq 5$, $x_1 \geq 0$, $x_2 \geq 0$. [8]
- If four points $2\vec{i} + 3\vec{j} - \vec{k}$, $\vec{i} - 2\vec{j} + 3\vec{k}$, $k\vec{i} + 4\vec{j} - 2\vec{k}$ and $\vec{i} - 6\vec{j} + 6\vec{k}$ are coplanar, find the value of k. [5]
 - If $\vec{a}'$, $\vec{b}'$, $\vec{c}'$ are reciprocal system of a, b, c. Prove that $(\vec{a} + \vec{b}) \cdot \vec{a}' + (\vec{b} + \vec{c}) \cdot \vec{b}' + (\vec{c} + \vec{a}) \cdot \vec{c}' = 3$. [5]
 - Find the radius of convergence and interval of convergence of the series $\sum_{n=1}^{\infty} \frac{(-1)^n (2x-1)^n}{n6^n}$. [5]

- Show that the infinite series $\sum_{n=1}^{\infty} \frac{1}{n^p}$: [5]
 - convergent if $p > 1$
 - divergent if $p \leq 1$
 - Find the condition of the line $ax + by + c = 0$ to be a tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Also find the point of contact. [5]
 - Sketch and describe the polar conic section: $r = \frac{6}{3 + \sin \theta}$. [5]

OR

Transform the equation $x^2 + 3xy + y^2 = 0$ through the point $(2, 3)$ and then through an angle of 45° .

- Find the length of perpendicular from the point $(1, 1, 1)$ to the line $\frac{x-2}{3} = \frac{y+2}{2} = \frac{z}{-1}$. Also find the foot of perpendicular and the equation of the perpendicular. [7]

OR

Prove that lines $\frac{x-2}{2} = \frac{y-4}{3} = \frac{z-6}{5}$ and $\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ intersects at a point. Also find the point of their intersection and the plane containing them.

- Find the shortest distance between the lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{1-y}{5} = \frac{z+2}{2}$. Also find the equation of S.D. [8]
- Find equation of the sphere through the circles $x^2 + y^2 = 4$, $z = 0$ and is cut by the plane $x + 2y + 2z = 0$ in a circle of radius 3. [7]
 - Find the equation of the cylinder whose generators are parallel to the line $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and guiding curve is $x^2 + y^2 = 16$, $z = 0$. [8]
- (Attempt All Questions) [4 x 2.5]
 - Show that $(\vec{a} \times \vec{b}) \cdot (\vec{a} \times \vec{c}) + (\vec{a} \cdot \vec{b})(\vec{a} \cdot \vec{c}) = (\vec{a} \cdot \vec{a})(\vec{b} \cdot \vec{c})$.
 - Find the linear relationship of the vectors $(1, 1, 1)$, $(1, 2, 3)$, $(3, 6, 9)$.
 - Find the centre and the radius of the sphere $x^2 + y^2 + z^2 + 2x - 4y - 6z + 5 = 0$.
 - Test series in Absolutely Convergent or not: $1 - \frac{1}{2\sqrt{2}} + \frac{1}{3\sqrt{3}} - \frac{1}{4\sqrt{4}} + \dots$

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Attempt all the questions.

- Check the given system of linear equations is consistent or not, if consistent then solve: $x + 2y + 3z = 1$, $2x + 3y + 2z = 2$, $2x + 3y + 4z = 1$. [7]
 - Let $V = \mathbb{R}^2$ be a vector space. Show that $W = \{(x, y) : x + 2y = 0\}$ is a vector subspace of V . [3]
 - Determine whether the vectors $(1, 2, 1), (2, 1, 0), (1, -1, 2)$ forms a basis for $\mathbb{R}^3$ or not? [5]
- Find the eigen values and the corresponding eigenvectors of the matrix $A = \begin{pmatrix} 2 & 1 & 2 \\ 0 & -1 & 3 \\ 0 & 1 & 1 \end{pmatrix}$. [7]

OR

Find the inverse of the given matrix $A = \begin{pmatrix} 4 & 3 & 1 \\ 2 & 1 & -2 \\ 1 & 2 & 1 \end{pmatrix}$ by using Cayley

Hamilton theorem.

- Using Simplex Method. Solve the following LPP: Maximize $Z = 3x_1 + 4x_2 + x_3$ subject to the constraints: $x_1 + 2x_2 + 3x_3 \leq 90$, $3x_1 + x_2 + 2x_3 \leq 80$, $2x_1 + x_2 + x_3 \leq 60$, $x_1, x_2, x_3 \geq 0$. [8]
- Define vector products of four vectors. Prove that: $[\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] = [\vec{a} \quad \vec{b} \quad \vec{c}]^2$. [5]
 - If $\vec{a} = (3, -4, 2)$, $\vec{b} = (1, 2, -3)$ and $\vec{c} = (2, 3, -1)$, then find the reciprocal vectors $\vec{a}'$ of a. [5]

c) Test the convergence and divergence of the series $\frac{1}{1+2} + \frac{2}{1+2^2} + \frac{3}{1+2^3} + \dots$ by using D'Alembert's ratio test. [5]

- Find the interval of convergence, centre of convergence and radius of convergence of the series $\sum_{n=1}^{\infty} \frac{10^{n+1}}{3^{2n}} x^n$. [7]
 - Show that the equation $16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$ represent a parabola. Also, find the equation of axis, vertex, equation of directrix and focus. [8]

OR

Sketch and describe the polar conic $r = \frac{10}{2+3 \sin \theta}$.

- Find the condition that the straight line $lx + my + n = 0$ will be tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Also find the point of contact. [7]
 - Prove that the lines $\frac{x+1}{1} = \frac{y-1}{2} = \frac{z+1}{3}$ and $-(x+2) = \frac{y-5}{2} = -z$ are coplanar. Also, find the point of contact and the equation of plane containing them. [8]
- Find the center and radius of the circle $x^2 + y^2 + z^2 + 12x - 12y - 16z + 111 = 0$, $2x + 2y + z = 17$. [7]
 - Plane through OX and OY include an angle α . Show that their lines of intersection lie on the cone $(z^2 + x^2)(y^2 + z^2) \cos^2 \alpha = x^2 y^2$. [4]
 - Find the equation of the cylinder with generators parallel to the line $\frac{x}{1} = \frac{y}{-2} = \frac{z}{3}$ and guiding curves is $x^2 + 2y^2 = 1, z = 0$. [4]
- (Attempt all the questions) [4 x 2.5]
 - Determine the rank of the matrix $A = \begin{pmatrix} 1 & -4 \\ -2 & 8 \end{pmatrix}$.
 - Show that the mapping $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x, x + y)$ is linear.
 - Find the volume of the parallelepiped whose concurrent edges are the vectors with position vectors $\vec{i} + 5\vec{j} + 3\vec{k}$, $\vec{i} + 2\vec{j} + 3\vec{k}$, $3\vec{i} + 7\vec{j} + 4\vec{k}$.
 - Show that the line $\frac{x-2}{2} = \frac{y-1}{3} = \frac{z-3}{-1}$ lies on the plane $2x - y + z - 6 = 0$.

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Attempt all the questions.

1. a) Check consistency and solve by Gauss elimination method: [7]

$$x + y + z = 6, x - y + z = 2, 2x + y - z = 1.$$

- b) Define eigen value and vector. Find the eigen value, eigenvector and diagonalize of the matrix $A = \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix}$ [8]

2. a) Solve the linear programming problem by Simplex method. [Constructing duality] [7]

$$\text{Minimize } Z = 2x_1 + 9x_2 + x_3 \text{ subject to } x_1 + 4x_2 + 2x_3 \geq 5, \quad 3x_1 + x_2 + 2x_3 \geq 4, \quad x_1 \geq 0, x_2 \geq 0, x_3 \geq 0.$$

OR

$$\text{Using simplex method maximize } Z = 150x_1 + 300x_2 \text{ subject to: } 2x_1 + x_2 \leq 16, \quad x_1 + x_2 \leq 8 \text{ and } x_1 \leq 3.5, \quad x_1 \geq 0, x_2 \geq 0.$$

$$\text{b) Solve (Big M-Method) Maximize } z = -3x_1 + 7x_2 \text{ subject to: } 2x_1 + 3x_2 \leq 5, \quad 5x_1 + 2x_2 \geq 3, \quad x_2 \leq 1. \quad [8]$$

3. a) State D'Alembert's Ratio test, show that: [7]

$$\text{i) } \sum_n \frac{1}{n} \text{ is divergent}$$

$$\text{ii) } \sum_n \frac{1}{n^2} \text{ is convergent}$$

$$\text{iii) } \sum_n \frac{(-1)^n}{n} \text{ is conditional divergent.}$$

- b) Find the center, radius and interval convergence of the power series:

$$\sum_{n=1}^{\infty} \frac{(x)^{2n+1}}{(-4)^n}.$$

4. a) Define the eccentricity of a conic section, and derive the equation of the ellipse in its standard form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. [7]

OR

Show that the line $lx + my + n = 0$ touches the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ if $a^2l^2 - b^2m^2 = n^2$. Also find the point of contact.

- b) Show the equations $16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$ represents a parabola. Also find the equations of axis and vertex. [8]

5. a) Define scalar triple product and reciprocal vectors. Find the reciprocal vector of: [8]

$$2\vec{i} + 3\vec{j} - \vec{k}, \quad \vec{i} - \vec{j} - 2\vec{k}, \quad -\vec{i} + 2\vec{j} + 2\vec{k}.$$

- b) Find the distance of the point (3, -4, 5) from the plane $2x + 5y + 6z = 16$ measured along a line with direction cosines proportional to 2, 1, -2. [7]

6. a) Find the shortest distance between the lines $\frac{x+2}{-4} = \frac{y}{1} = \frac{z-7}{1}$ and $\frac{x+3}{-4} = \frac{y-6}{3} = \frac{z}{2}$. Also find the equation of shortest distance. [7]

- b) Find the equation of sphere for which circle $x^2 + y^2 + z^2 + 7y - 2z + 2 = 0, \quad 2x + 3y + 4z = 8$ is a great circle. [8]

7. Attempt all the questions. [2.5 x 4]

$$\text{a) Show that } [\vec{a} \times \vec{b} \quad \vec{b} \times \vec{c} \quad \vec{c} \times \vec{a}] = [\vec{a} \quad \vec{b} \quad \vec{c}]^2.$$

- b) Find the centre of the hyperbola:

$$5x^2 - 4y^2 + 20x + 8y = 4.$$

- c) Show that the mapping. $T: R^2 \rightarrow R^2$ define by $T(x, y) = (x, x + y)$ is linear.

- d) Plane through the OX and OY include an angle α . Show that their lines of intersection lie on the cone is $(z^2 + x^2)(y^2 + z^2) \cos^2 \alpha = x^2 y^2$.

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Attempt all the questions.

- Check the given system of linear equations is consistent or not. If consistent then solve: [8]
 $x + 2y - 3z = 9, \quad 2x - y + 2z = -8, \quad 3x - y - 4z = 3.$
 - Verify Cayley Hamilton theorem for the matrix [7]

$$A = \begin{pmatrix} 1 & 0 & 2 \\ 0 & -1 & 1 \\ 0 & 1 & 0 \end{pmatrix}$$
- Define basis for a vector space. Show that the vectors (1, 2, 1), (2, 1, 0), (1, -1, 2) for a basis of $\mathbb{R}^3$. [7]
 - Find Eigen value and corresponding Eigen vectors of the matrix. [8]

$$A = \begin{pmatrix} 3 & 1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{pmatrix}$$

OR

Define Rank of matrix. Find Rank of following matrix by using Echelon form:

$$\begin{pmatrix} 2 & 1 & 3 \\ 4 & 7 & 13 \\ 4 & -3 & -1 \end{pmatrix}$$

- Using simplex method, solve the following LPP: [7]
 Maximize $Z = 2x_1 - x_2 + 2x_3$ subject to constraints
 $x_1 + 2x_2 - 2x_3 \leq 20, \quad 2x_1 + x_2 \leq 10, \quad x_2 + 2x_3 \leq 5$ and $x_1, x_2, x_3 \geq 0$

OR

Construct the dual problem and solve by simplex method:

Minimize $Z = 8x_1 + 9x_2$ subject to constraints
 $x_1 + x_2 \geq 6, \quad 3x_1 + x_2 \geq 21$ and $x_1, x_2 \geq 0$

- State and prove Leibnitz's theorem for convergence of an infinite series. [4]
- Test the convergence and divergence of the following series by using D'Alembert ratio test: [4]

$$\frac{1}{2} + \frac{4}{2^2} + \frac{9}{2^3} + \dots$$
- Define scalar and vector product three vectors. Derive the expression for vector product of three vectors. [8]
 - Find the set of reciprocal vector of $a = 2i + 3j - k$, $b = i - j - 2k$ and $c = -i + 2j + 2k$. [7]
- Define conic section and derive the standard equation of Ellipse. [8]
 - Obtain the condition that the line $lx + my + n = 0$ may touch the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. [7]
- Show that the lines $\frac{x+4}{3} = \frac{y+6}{5} = \frac{1-z}{2}$ and $3x - 2y + z + 5 = 0 = 2x + 3y + 4z$ are coplanar. Find their point of intersection and the plane in which they lie. [7]
 - Find the centre and radius of circle $x^2 + y^2 + z^2 + 12x - 13y - 16z + 111 = 0, \quad 2x + 2y + z = 17$. [8]
- Attempt all: (2 x 5)
 - Find the equation of the sphere one of whose great circle is:
 $x^2 + y^2 + z^2 = 4, \quad x^2 + y^2 + z^2 - 2x + 4y - 6z = 11$
 - Show that the points (4, 5, 1), (3, 9, 4), (-4, 4, 4) and (0, -1, -1) are coplanar.

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Attempt all the questions.

1. a) Check the given system of linear equations is consistent or not, if consistent then solve: [7].
 $x + y + z = 1, \quad x + 2y + 3z = -4, \quad x + 3y + 7z = 13.$

- b) Find eigenvalue and eigenvector of the matrix, $\begin{bmatrix} 2 & 1 & 2 \\ 0 & -1 & 3 \\ 0 & 1 & 1 \end{bmatrix}$. [8]

2. a) i. Let $V = \mathbb{R}^2$ be a vector space, show that $W = \{(x, y) : x + 2y = 0\}$ is a vector sub-space of V . [3]
 ii. Check the vectors $(1, 2, 1), (2, 1, 0), (1, -1, 2)$ for a basis for $\mathbb{R}^3$ or not. [4]

- b) Solve the given LPP by using appropriate simplex method. [8]

$$\text{Max } Z = 2x_1 + x_2$$

$$\text{S.t. } 2x_1 - x_2 \geq 2; \quad x_1 - x_2 \leq 2; \quad x_1 + x_2 \leq 4; \quad x_1 \geq 0, x_2 \geq 0.$$

OR

Construct the dual problem and solve by simplex method.

$$\text{Minimize } Z = 8x_1 + 9x_2 \text{ subject to constraints } x_1 + x_2 \geq 6, \quad 3x_1 + x_2 \geq 21, \quad x_1, x_2 \geq 0.$$

3. a) State and prove p-series test. [7]
 b) Find the interval and radius of convergence of the series, [8]

$$\sum_{n=0}^{\infty} \frac{(x-2)^n}{10^n}.$$

4. a) Find the value of λ when the line $y = 2x + \lambda$ is tangent to the hyperbola $3x^2 - y^2 = 3$. [7]
 b) Show that the equations $9x^2 - 24xy + 16y^2 - 50x - 100y + 225 = 0$ represents a parabola. Also, determine its equation of axis, vertex, latus rectum, focus and directrix. [8]

5. a) Examine, whether the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-4}{4}$ and $\frac{x-2}{3} = \frac{y-3}{4} = \frac{z-4}{5}$ are coplanar or not. If so, find the equation of plane containing them. [7]

OR

Find the length and equation of the shortest distance between the lines

$$\frac{x}{2} = \frac{y}{-3} = \frac{z}{1} \quad \text{and} \quad \frac{x-2}{3} = \frac{1-y}{5} = \frac{z+2}{2}.$$

- b) Find the centre and radius of circle $x^2 + y^2 + z^2 + 12x - 13y - 16z + 111 = 0, \quad 2x + 2y + z = 17$. [8]

6. a) Find the equation of right circular cylinder whose guiding curve is $x^2 + y^2 + z^2 - x - y - z = 0, \quad x + y + z = 1$. [7]
 b) i. If a, b, c be any three vectors. Show that $[a+b \quad b+c \quad c+a] = 2[a \quad b \quad c]$. [4]
 ii. Define reciprocal system of vectors. Find the reciprocal of a for $a = (-1, 2, 2), b = (2, 3, 1)$ and $c = (1, -1, -2)$. [4]

7. Attempt all the questions. [2.5 x 4]

- a. Find the area of the triangle whose vertices are $(-1, 1, 2), (2, 5, -1), (1, 2, 3)$.
 b. Define cross product of four vectors.
 c. Find the equation of plane through the origin and having direction cosines proportional to $(5, 2, -3)$ and $(2, 1, -2)$
 d. For what value of 'k' the plane $x - 2y - 2z = k$ touches the sphere $x^2 + y^2 + z^2 + 4y - 2x - 6z = -5$.

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Attempt all the questions.

1. a) Check whether the given system of linear equations is consistent or not, if consistent then solve the system of linear equations: [7]
 $2x + 5y + 6z = 13, \quad 3x + y - 6z = 13, \quad x - 3y - 8z = -13.$
 b) [4+4]
 i) State Cayley Hamilton theorem. Verify it for the matrix $A = \begin{pmatrix} 1 & -2 \\ 4 & 5 \end{pmatrix}$.
 ii) Check whether the set of vectors $\{(1,1,0), (1,0,1), (3,1,1)\}$ for a basis for $\mathbb{R}^3$ or not.
2. a) Define eigen value and eigen vector. Find the eigen values and corresponding eigen vectors of $\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$. [7]
 b) Convert the following primal LPP into dual LPP and solve by using simplex method: [8]
 Minimize $Z = 8x_1 + 9x_2$ subject to $x_1 + 3x_2 \geq 4, \quad 2x_1 + x_2 \geq 5, \quad x_1 \geq 0, x_2 \geq 0.$
3. a) Find the set of reciprocal system of vectors to $a = 2i + 3j - 2k, b = i - j - 2k$ and $c = -i + 2j + 2k$. [7]
 b) Find the interval of convergence, centre of convergence and radius of convergence of an infinite series $\sum \frac{n^2}{2^n} (x + 4)^n$. [8]
4. a) State Cauchy root test. Test the convergence of the following infinite series: [7]
 i. $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots$

ii. $\sum \left(\frac{2n}{n+1} \right)^n$

- b) Find the vertex, eccentricity, foci, and equation of directrix of the ellipse: $3x^2 + 4y^2 - 12x - 8y + 4 = 0$. [8]
5. a) Find the condition for the line $y = mx + c$ to be the tangent to the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$. Also, find the point of contact. [7]

OR

Sketch and describe the polar conic $r = \frac{12}{3+2 \cos \theta}$.

- b) Define skew lines. Find the shortest distance between the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-2}{3} = \frac{y-4}{4} = \frac{z-5}{5}$. Also, find the equation of the line of shortest distance. [8]
6. a)
 i. Find the equation of the cone with vertex at the origin and which passes through the curve of intersection of $ax^2 + by^2 + cz^2 = 1$ and $lx + my + nz = p$. [2]
 ii. Find the equation of the right circular cylinder of radius 2 whose axis is the line $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{2}$. [5]
 b) Find the centre and radius of the circle $x^2 + y^2 + z^2 - 8x + 4y + 8z - 45 = 0, \quad x - 2y + 3z = 3$. [8]
- OR**
- Show that the plane $2x - 2y + z = -12$ touches the sphere $x^2 + y^2 + z^2 - 2x - 4y + 2z = 3$. Also find the point of contact.
7. Attempt all the questions: [2.5 x 4]
 a. Check whether the transform $T: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $T(x, y) = x + y$ is linear or not.
 b. Find the volume of the parallelepiped whose concurrent edges are given by: $\vec{a} = \vec{i} + 2\vec{j} + 3\vec{k}, \vec{b} = 3\vec{i} + 4\vec{j} - 5\vec{k}, \vec{c} = \vec{i} - 2\vec{j} + 3\vec{k}$.
 c. Transform the equation $x^2 + 3xy + y^2 = 0$ in which the origin is transformed to (2, 3) with axes remaining parallel to the old axes.
 d. Show that the line joining the points (-2, 1, 3) and (1, -3, 4) is parallel to the plane $2x + 3y + 6z + 5 = 0$.